

BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE EXAMINATION, 2024 ONWARDS

Time: 20 minutes **PHYSICS PAPER – I MODEL QUESTION PAPER**

Max. Marks: 17

(Science Groups)

SECTION 'A'

(MULTIPLE CHOICE QUESTIONS) – (M.C.Qs.)

(Marks : 17)

NOTE: i) This section consists of 17 part questions and all are to be answered. Each question carries one mark.

ii) The correct answer bubble must be filled on OMR sheet 1) pasted in answer script.

iii) Use only blue / black ball point pen or pointer on OMR sheet.

iv) Avoid using pencil / White-o pen on OMR sheet.

1. Select the correct answer for each from the given options:

- 1) Unit of solid angle is:
A) second B) kilogram **C) steradian** D) candela
- 2) The velocity of a particle at an instant is 10 m/s and after 5 second the velocity of particle is 20m/s. The velocity 3 second before the initial instant is:
A) 8 m/s **B) 4 m/s** C) 6 m/s D) 7 m/s
- 3) If momentum is increased by 20% then K.E increases by:
A) 44% B) 55% C) 66% D) 77%
- 4) A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms:
A) his angular velocity increases B) his angular velocity remains same
C) his rotational inertia decreases **D) his angular momentum remains the same**
- 5) The absolute potential energy of an object depends on:
A) The object's mass and height from earth's surface B) The object's mass and speed
C) The object's shape and size D) The object's colour and temperature
- 6) One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by ΔP the pressure at the larger piston:
A) increases by $2\Delta P$ B) increases by $\Delta P/2$
C) increases by ΔP D) increases by $4\Delta P$
- 7) A sky diver falls through the air with terminal velocity. The force of air resistance on him is:
A) half of his weight **B) equal to his weight**
C) twice his weight D) cannot be determined from the information given
- 8) The force between two charges placed in air is F , if air is replaced by a medium of relative permittivity ϵ_r then force is reduced to:
A) $F \epsilon_r$ B) $\frac{F}{\epsilon_r}$ C) $\frac{\epsilon_r}{F}$ D) $\epsilon \epsilon_r$
- 9) The charging of a capacitor through a resistance follows:
A) linear law B) square law **C) exponential law** D) inverse square law
- 10) A wire of uniform area of cross-section A , length L and resistance R is cut into two parts. The resistivity of each part:
A) becomes zero B) is halved C) is doubled **D) remain same**
- 11) A heat-sensitive device whose resistivity changes with the change in temperature is called:
A) conductor B) resistor **C) thermistor** D) thermometer
- 12) A child swinging on a swing in sitting position, stands up, then the time period of the swing will:
A) Increase **B) decrease**
C) remains the same D) increases if the child is long and decreases if the child is short
- 13) A heavily damped system has a fairly flat resonance curve in:
A) An acceleration time graph **B) An amplitude frequency graph**
C) Velocity time graph D) Distance-time graph
- 14) If v_a , v_h and v_m are the speeds of sound in air, hydrogen gas and a metal at the same temperature, then: ,
A) $v_a > v_h > v_m$ **B) $v_m > v_h > v_a$** C) $v_h > v_m > v_a$ D) $v_h > v_m > v_a$
- 15) In Young's double slit experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be:
A) 4 times B) 1/4 times C) doubled D) unchanged
- 16) A hill separates a television (TV) transmitter from a house. The Transmitter cannot be seen from the house but still the TV in the house has good reception. What wave phenomena make it possible:
A) Coherence of waves **B) Diffraction of waves**
C) Interference of waves D) Refraction of waves
- 17) The process of superimposing signal frequency on carrier wave is known as:
A) Transmission B) Detection C) Reception **D) Modulation**